Pierre-Amaury GRUMIAUX

Contact

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Nantes, France

Hard Skills

Machine learning and deep learning
- PyTorch, Tensorflow
- Model optimization: quantization, compression, ONNX, TFLite

Audio & signal processing
- Musical acoustics, psychoacoustics, spatial audio
- Audio augmentation, feature extraction

MLOps & development
- Production deployment & embedded systems optimization
- Model evaluation and custom metrics
- CI/CD pipelines

Programming tools
- Python (signal and audio libraries)
- C++, Matlab, Faust
- Git, Linux, cloud computing (AWS)

Music
- Piano (+25 years), music theory
- DAWs, synthesizers, sound design, mixing, mastering

Soft Skills

- Technical leadership | Autonomous | Problem-solver
- Cross-functional collaboration | Rigorous | Strategic thinking

Education

2018 – 2021 **PhD degree**
GIPSA-lab, Grenoble

2017 – 2018 **Research master degree**
Acoustics, audio signal processing and computer music (ATIAM)
Sorbonne Universités, Paris

2013 – 2017 **Engineering degree**
Computer science
Centrale Lille, Lille

2011 – 2013 **Preparatory classes**
Lycée Marcelin Berthelot, Saint-Maur

Hobbies

Volleyball (national level competitions), music making, science & history books, museums, chess, nature exploration, hiking,

About Me

Audio ML Engineer with a strong research foundation (PhD, postdoc) and practical experience shipping deep learning models for real-time audio applications on embedded devices. I'm driven by the convergence of AI and audio/music technology, and I thrive in environments where I can push both research boundaries and production quality. My 25+ years as a pianist and my work in music production fuel my passion for this field.

Professional Experience

Sonaid | Audio ML engineer
2024 – present

- Built end-to-end deep learning pipeline from scratch: dataset creation with custom annotations and audio augmentation strategies, model architecture, training, and evaluation with custom metrics
- Researched and implemented state-of-the-art audio processing techniques for audio event detection
- Optimized models for embedded systems through compression, quantization, latency reduction, and memory optimization while balancing accuracy vs. computational constraints
- Prepared and deployed models for production environments
- Led technical strategy: defined AI roadmap, prioritized features, and integrated customer feedback into model improvements
- Collaborated cross-functionally with product, hardware, and backend teams while maintaining technical documentation

LS2N, Centrale Nantes | Postdoctoral researcher
2022 – 2023

Bandwitch extension of musical signals with differentiable models
- State-of-the-art review of bandwidth extension methods
- Design and evaluation of several DDSP models for monophonic and polyphonic bandwidth extension

Orange Labs & GIPSA-lab | PhD researcher
2018 – 2021

Deep learning for speaker counting and localization with Ambisonics signals
- State-of-the-art review of sound source counting and localization
- Preparation of datasets including multichannel multi-source speech signals with reverberation and noise in Ambisonics format
- Deep learning models for speaker counting, speaker localization and hybrid methods : CNN, RNN, CRNN, ResNet, self-attention
- Investigations regarding a novel 3D audio representation

IRCAM | Research intern
Feb. – Jul. 2018

Automatic drums transcription with deep learning

Audionamix | Research intern
Apr. – Aug. 2017

Audio-to-lyrics alignment for polyphonic music

CCRMA & Mines ParisTech | Research intern
Jun. – Aug. 2016

Physical modeling based sound synthesis with Faust